

LIBERAL, KS, USA

WMO: 724516

Lat: 37.050N

Lon: 100.967W

Elev: 876

StdP: 91.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23020

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.3	-17.9	0.9	-10.2	-16.2	1.0	-7.7	15.5	7.0	13.9	5.2	4.6	20	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	37.9	20.8	37.1	20.8	35.5	20.5	23.5	31.9	22.8	31.7	22.1	31.3	7.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	17.8	25.9	20.9	17.3	25.6	19.4	15.8	25.1	74.6	31.5	71.8	31.9	69.3	31.3	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.6	11.3	-18.5	40.7	2.6	1.4	-20.5	41.7	-22.0	42.5	-23.5	43.3	-25.4	44.3
			-19.5	24.4	2.3	0.6	-21.1	24.8	-22.4	25.2	-23.7	25.5	-25.4	25.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.3	2.9	7.8	12.7	18.3	24.2	26.8	25.7	21.4	14.0	6.9	1.3	(6)
(7)		DBStd	10.37	5.45	6.11	5.89	5.30	4.72	3.62	2.85	2.98	4.44	5.17	5.05	5.33	(7)
(8)		HDD10.0	1032	271	208	110	32	2	0	0	1	21	117	270		(8)
(9)		HDD18.3	2574	527	433	328	179	60	3	0	1	22	151	342	527	(9)
(10)		CDD10.0	2369	2	8	41	113	260	425	519	485	344	145	25	2	(10)
(11)		CDD18.3	870	0	0	2	10	59	178	262	227	116	17	0	0	(11)
(12)		CDH23.3	10938	1	9	72	274	853	2175	3285	2615	1336	295	23	0	(12)
(13)	CDH26.7	5739	0	1	15	94	378	1154	1903	1444	643	105	2	0	(13)	
(14)	Wind	WSAvg	5.5	5.0	5.4	5.9	6.4	5.9	6.0	5.3	4.9	5.4	5.5	5.3	5.0	(14)
(15)	Precipitation	PrecAvg	499	12	15	33	37	72	75	74	58	49	38	22	13	(15)
(16)		PrecMax	867	50	59	229	108	224	205	301	172	162	270	91	48	(16)
(17)		PrecMin	308	0	0	0	1	1	4	5	3	0	0	0	0	(17)
(18)		PrecStd	115	11	16	41	28	48	49	52	39	37	47	24	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	24.2	28.7	32.5	36.4	40.0	39.8	38.9	37.5	32.7	26.3	20.2	(19)
(20)			MCWB	9.6	10.5	13.6	14.7	17.5	19.4	21.5	21.0	19.8	17.0	12.6	10.0	(20)
(21)		2%	DB	17.5	21.0	25.6	29.1	33.0	37.2	37.9	37.5	35.0	29.6	22.6	16.3	(21)
(22)			MCWB	7.8	9.6	12.6	13.8	17.5	20.1	21.2	21.1	19.4	16.4	11.6	7.8	(22)
(23)		5%	DB	13.9	17.2	22.2	27.0	31.2	35.1	37.2	36.1	32.7	27.1	19.2	13.1	(23)
(24)			MCWB	6.4	8.3	11.5	14.2	17.6	20.3	21.2	21.1	19.4	15.5	10.2	6.7	(24)
(25)		10%	DB	11.0	13.0	18.8	23.0	28.0	32.9	35.8	33.9	30.9	23.6	16.4	10.8	(25)
(26)			MCWB	5.0	6.1	10.3	12.8	17.0	20.2	21.2	21.0	19.1	14.4	8.8	5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	12.6	16.5	18.2	21.5	24.0	24.6	24.0	22.3	20.1	14.7	11.7	(27)
(28)			MCDWB	19.1	19.8	23.9	24.1	28.9	31.0	32.5	32.3	30.9	26.6	21.4	16.3	(28)
(29)		2%	WB	8.5	10.4	14.3	16.6	20.4	22.9	23.7	23.1	21.4	18.7	12.7	9.2	(29)
(30)			MCDWB	16.5	19.3	21.0	24.0	27.6	31.3	32.1	32.2	30.2	24.0	20.0	15.8	(30)
(31)		5%	WB	7.0	8.9	12.7	15.1	19.4	22.0	22.9	22.5	20.7	17.4	11.2	7.2	(31)
(32)			MCDWB	13.7	17.1	19.9	23.7	26.9	31.0	32.0	31.6	29.6	23.3	18.7	12.9	(32)
(33)		10%	WB	5.4	7.0	11.1	13.9	18.3	21.3	22.3	21.8	20.0	15.8	9.7	5.5	(33)
(34)			MCDWB	10.8	13.0	18.9	22.5	25.9	30.1	31.6	30.8	28.3	22.6	15.9	10.7	(34)
(35)	Mean Daily Temperature Range	MDBR	14.7	15.1	15.9	15.6	14.9	14.6	14.6	14.1	14.6	15.1	15.5	13.9	(35)	
(36)		5% DB	MCDBR	20.5	21.5	21.4	21.1	19.0	17.9	17.1	17.3	17.7	19.6	20.5	18.9	(36)
(37)			MCWBR	12.4	12.3	11.1	9.6	7.7	6.1	5.2	5.2	6.2	9.0	11.2	11.7	(37)
(38)		5% WB	MCDWB	19.7	20.3	18.4	17.6	16.1	15.4	14.8	14.7	14.9	15.3	18.6	18.0	(38)
(39)			MCWBR	12.1	11.8	10.2	8.8	7.4	6.3	5.0	5.3	5.7	7.9	10.5	11.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.267	0.298	0.327	0.345	0.374	0.384	0.368	0.338	0.303	0.272	0.258	(40)	
(41)		taud	2.569	2.561	2.474	2.414	2.430	2.392	2.395	2.408	2.474	2.543	2.565	2.572	(41)	
(42)		Ebn at Noon	954	984	972	954	935	904	891	901	915	925	927	932	(42)	
(43)		Edh at Noon	74	84	102	114	115	119	118	114	101	85	73	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.79	3.60	4.85	6.00	6.88	7.45	7.28	6.41	5.43	4.03	3.05	2.46	(44)	
(45)		RadStd	0.25	0.37	0.41	0.34	0.42	0.39	0.31	0.32	0.28	0.46	0.28	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page